

RESEARCH ARTICLE

High-Efficiency Bifacial Perovskite Solar Cell With Optimization of Tilt Angle and Albedo

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ABSTRACT

This study develops an integrated computational framework for optimizing a bifacial perovskite solar cell (PSC) with $\text{MgF}_2/\text{FTO}/\text{CdS}/\text{Ca}_3\text{N}_2/\text{MgF}_2$ architecture by combining density functional theory (DFT), SCAPS-1D simulations, OghmaNano modeling, and a mathematical performance model. The proposed device architecture is structurally novel, with MgF_2 introduced as an anti-reflective coating, enhancing light in-coupling and improving overall optical performance. DFT calculations provided the essential electronic and optical parameters required for realistic device construction, while SCAPS-1D was used to evaluate photovoltaic behavior. Under standard conditions, the optimized PSC achieved a J_{sc} of 43.79 mA/cm^2 , V_{oc} of 0.91 V , FF of 83.50% , power conversion efficiency of 33.06% , and bifacial factor of 76.245% . To ensure robustness, the device was additionally simulated using OghmaNano, whose finite-element drift-diffusion and transfer-matrix optical solvers produced closely matching performance values. This strong agreement provides an independent numerical backbone that validates the SCAPS results and confirms the physical reliability of the proposed design. A mathematical model was further used to assess outdoor operation, revealing an optimal tilt angle of $\sim 35^\circ$ and significant rear-side gains on high-albedo surfaces such as fresh snow ($\alpha = 0.8$). Overall, this combined DFT-SCAPS-OghmaNano-mathematical methodology is novel and offers a validated and comprehensive route for designing high-efficiency bifacial PSCs under practical conditions.

1 | Introduction

Solar energy has emerged as a leading renewable energy source owing to its abundance, sustainability, and potential to meet the growing global energy demand [1, 2]. Among various photovoltaic (PV) technologies, perovskite solar cells (PSCs) have gained exceptional attention due to their remarkable power conversion efficiency (PCE), low fabrication cost, and compatibility with flexible substrates [3–5]. Although the earliest lead-halide PSCs demonstrated a modest efficiency of 3.8% , continuous advancements in material design, interface engineering, and device architecture have elevated their efficiency to an impressive 25.7% over the past decade [6, 7].

Traditional perovskites generally adopt the ABX_3 crystal structure, where A represents an organic or inorganic cation (e.g., MA^+ , FA^+ , Cs^+), B is a divalent metal cation (Pb^{2+} , Sn^{2+}), and X denotes a halide ion (I^- , Br^- , Cl^-) [8]. Beyond these, several alternative thin-film materials such as CIGS ($\text{Cu}(\text{In,Ga})\text{Se}_2$), CdTe, and kesterite ($\text{Cu}_2\text{ZnSn}(\text{S,Se})_4$) have also been widely studied for their high optical absorption and long-term operational stability [9, 10]. However, the toxicity and instability of lead-based perovskites have driven intensive research toward lead-free alternatives that maintain comparable efficiency and improved environmental compatibility [11–13]. In this context, lead-free perovskites such as RbGeI_3 , Cs_2SnI_6 , and CsSnCl_3 have been explored, yet they suffer from critical limitations including

poor ambient stability, susceptibility to oxidation (particularly $\text{Sn}^{2+} \rightarrow \text{Sn}^{4+}$ conversion), deep-level defect formation, and inferior charge-transport properties [14, 15]. These shortcomings hinder their large-scale deployment and long-term reliability, emphasizing the need for more stable and efficient lead-free systems.

Furthermore, achieving practical commercialization of PSCs requires not only high efficiency but also robust environmental stability, particularly under conditions of humidity, temperature fluctuations, and prolonged solar exposure [16, 17]. In recent years, bifacial PSCs, capable of harvesting sunlight from both front and rear surfaces, have emerged as a powerful strategy to enhance overall energy output [18, 19]. By utilizing direct, diffuse, and reflected (albedo) light, bifacial devices deliver significantly higher energy yields than their monofacial counterparts [20]. This dual-sided absorption mechanism, combined with sensitivity to environmental reflectance, makes bifacial perovskite devices especially promising for real-world outdoor applications, where varying albedo surfaces strongly influence rear-side power generation [21]. Consequently, bifacial PSCs represent a forward-looking direction toward maximizing solar energy utilization and ensuring sustainable, high-efficiency photovoltaic operation.

In bifacial PSCs, achieving the highest PCE for both the front and rear sides requires a specific tilt angle to capture both direct and diffuse sunlight effectively [22]. This unique characteristic of bifacial PSCs allows for improved energy collection, which is crucial for enhancing overall performance [23]. To optimize the utilization of the bifacial property of PSCs, a mathematical approach is essential to determine the optimal tilt angle that maximizes the overall energy conversion efficiency. Without such a model, the full potential of bifacial PSCs cannot be efficiently harnessed. Understanding the impact of tilt angle on both the front and rear PCE can help in determining the best configuration for different environmental conditions, ultimately increasing the practical viability of bifacial PSCs in diverse real-world settings [24]. The performance of bifacial PSCs is also highly sensitive to the albedo of surrounding surfaces, including grass, concrete, snow, and water, which affect the amount of diffuse and reflected light incident on the rear side [25]. While several works have explored front-side PCE optimization, systematic studies considering both front and rear efficiencies under varying tilt angles and environmental conditions remain limited [26, 27].

In recent years, extensive research efforts have been devoted to improving photovoltaic device performance through material engineering, device architecture optimization, and numerical modeling. Several studies have focused on understanding material growth and stability, such as the fabrication of MoOx nanobelts via annealing of sputtered Mo films, where annealing duration was shown to strongly influence phase formation, crystallinity, and electrical properties, with MoO₃ emerging as the dominant phase [28]. Parallel efforts in thin-film photovoltaics have explored alternative buffer layers for CZTS solar cells, identifying ZnS as a promising replacement for CdS and demonstrating the critical role of absorber and buffer thickness optimization under temperature variations [29]. Temperature-dependent performance has also been systematically studied in conventional photovoltaic technologies, revealing that while

CdTe exhibits lower thermal sensitivity, Si and GaAs retain higher efficiencies under standard operating conditions [30].

Beyond conventional systems, long-term stability has emerged as a key challenge for perovskite solar cells, with multi-year storage studies revealing severe efficiency degradation driven by perovskite decomposition, ion migration, and Pb diffusion toward metal contacts [31]. To address performance enhancement from a device-architecture perspective, bifacial perovskite solar cells have been numerically investigated using SCAPS-1D, demonstrating that transitioning from monofacial to bifacial designs significantly enhances energy harvesting, with reported bifacial factors exceeding 70% [32]. Complementary computational studies have examined the role of hole transport materials in $\text{CH}_3\text{NH}_3\text{PbI}_3$ -based devices, predicting efficiencies above 25% using Cu_2O HTMs and highlighting the potential of HTM-free architectures for cost-effective and stable operation [33]. In addition, emerging absorber materials such as MoS_2 have been numerically assessed, where optimized ETL/HTL combinations enabled power conversion efficiencies exceeding 20%, underscoring the importance of transport-layer engineering [34]. Collectively, these studies demonstrate that advances in material selection, interface optimization, bifacial architectures, and multi-physics simulation frameworks are central to achieving high-efficiency, stable, and scalable next-generation solar cells.

In this study, a bifacial PSC structure was developed with layers of $\text{MgF}_2/\text{FTO}/\text{CdS}/\text{Ca}_3\text{Ni}_2/\text{MgF}_2$. Ca_3Ni_2 is an emerging lead-free perovskite material that has attracted attention due to its favorable optoelectronic properties and environmental friendliness. It exhibits a suitable bandgap for optoelectronic applications, providing effective absorption of high-energy photons. Importantly, Ca_3Ni_2 demonstrates excellent thermal and chemical stability compared to many conventional perovskites [35], which often suffer from degradation under operational conditions. Its non-toxic composition further enhances its appeal as a sustainable alternative for photovoltaic and optoelectronic devices, making it a promising candidate for future energy applications. The MgF_2 layer is an anti-reflective coating, minimizing sunlight reflection and thus enhancing light absorption. To gain insights into the material properties, first-principles density functional theory (DFT) was employed to calculate the bandgaps of the materials used. The calculated bandgaps were 3.184 eV for FTO, 1.64 eV for CdS, and 2.20 eV for Ca_3Ni_2 . These values of bandgaps reveals that they are highly suitable for designing high-efficiency PSC. In addition, DFT was utilized to compute the optical properties of MgF_2 and Ca_3Ni_2 , which are essential for understanding the interaction of light with the material layers. To further optimize the performance of the device, the SCAPS simulation tool was employed to simulate and fine-tune the photovoltaic parameters. The optimized results show a short-circuit current density (J_{sc}) of 43.79 mA/cm^2 , an open-circuit voltage (V_{oc}) of 0.91 V, a fill factor (FF) of 83.50%, and a PCE of 33.06% at 300 K. The front PCE was calculated to be 33.06%, while the rear PCE was 25.21%, resulting in a bifacial factor of 76.245%.

To validate the robustness of the SCAPS-derived results, the optimized device structure was further analyzed using the OghmaNano simulation platform. OghmaNano employs a finite-element drift-diffusion solver coupled with transfer-matrix optical modeling, offering a more detailed description of carrier

transport and light propagation than conventional 1D solvers. The J - V characteristics obtained from OghmaNano produced performance parameters closely matching those from SCAPS, with comparable other PV metrics. This strong agreement between two independent numerical frameworks confirms that the high PCE predicted by SCAPS is not an artefact of a single modeling approach. Instead, the OghmaNano results provide a solid computational backbone, reinforcing the physical reliability and stability of the $\text{MgF}_2/\text{FTO}/\text{CdS}/\text{Ca}_3\text{Ni}_2/\text{MgF}_2$ architecture and validating the feasibility of the proposed device design under standard operating conditions. Following these simulations, a mathematical model was developed to explore the effects of tilt angles on the front and rear PCE for various surface types, including fresh snow, concrete, grass, and water, with different albedo values (0.8, 0.4, 0.25, and 0.06) [36]. This work introduces novel insights into optimizing the bifacial performance of PSCs, offering a comprehensive framework for enhancing energy harvest across various environmental conditions and surface types.

The novelty of this work lies in being, to the best of our knowledge, the first comprehensive study to integrate four complementary methodologies-Density Functional Theory (DFT), SCAPS-1D simulation, OghmaNano drift-diffusion modeling, and a mathematical angular model-within a unified, multi-scale framework for bifacial PSCs. Previous studies have generally focused either on material-level DFT analysis or device-level SCAPS modeling, without establishing a direct, data-driven connection between these domains or validating results using an independent solver. In contrast, this work introduces a cross-domain workflow that translates first-principles insights into SCAPS-based device optimization and further confirms their reliability through OghmaNano verification. The subsequent integration of a mathematical angular model bridges theoretical design, device feasibility, and real-world operational performance. Furthermore, the dual MgF_2 layers incorporated on both sides of the $\text{MgF}_2/\text{FTO}/\text{CdS}/\text{Ca}_3\text{Ni}_2/\text{MgF}_2$ architecture act as multifunctional coatings, serving simultaneously as anti-reflective and optical modulation layers. This design ensures enhanced optical adaptability and efficiency retention at practical tilt angles (30° – 35°), where bifacial modules are typically deployed in real-world conditions. By addressing both intrinsic material properties and extrinsic operational parameters, this study presents a holistic and physically consistent pathway toward stable, high-performing bifacial PSCs under realistic outdoor illumination. Additionally, the selection of Ca_3Ni_2 as the absorber layer further reinforces the novelty, offering a non-toxic, lead-free alternative with favorable optoelectronic properties for the application of bifacial PSC. Collectively, these features establish this work as a first-of-its-kind, fully integrated theoretical and mathematical model that advances the frontier of sustainable bifacial PSC design.

2 | Device Simulation and Mathematical Model Information

In this study, first-principles DFT calculations were conducted using the CASTEP software to assess the material properties and optimize the performance of the bifacial PSC [37, 38]. CASTEP, developed by Materials Design Inc., is particularly

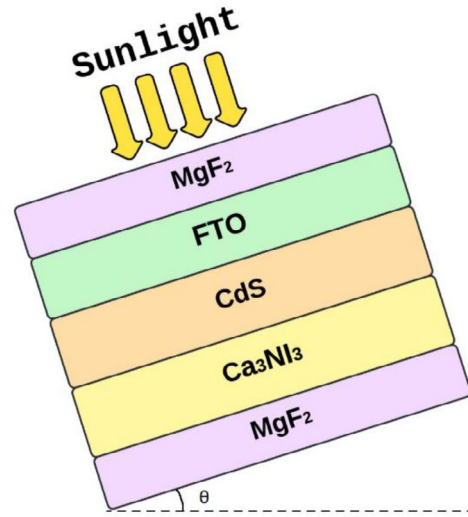


FIGURE 1 | Architecture of proposed bifacial PSC.

effective for first-principles DFT calculations, enabling precise evaluation of material properties such as bandgaps and optical characteristics [39, 40]. The DFT simulations were performed using the HSE06 hybrid functional to improve bandgap accuracy [13]. Additionally, appropriate norm-conserving pseudopotentials were employed throughout the calculations to ensure the reliability of our results [41]. The bandgaps of the materials used in the device were calculated using DFT, and six key optical properties were derived from CASTEP, providing insights into the interaction of light with the material layers [42]. These optical properties were essential for optimizing the PCE of the device [43, 44].

Additionally, the SCAPS-1D simulation tool, developed by the University of Gent (Belgium), was employed to further optimize the photovoltaic parameters [45, 46]. SCAPS-1D is particularly suited for simulating photovoltaic devices, offering detailed modeling of layer interactions and optimization of key parameters like J_{sc} , V_{oc} , FF, and PCE [47, 48]. This tool enabled the accurate modeling of the bifacial PSC structure, accounting for interactions between layers such as FTO, CdS, and Ca_3Ni_2 . The device structure is represented in Figure 1. The Tables S1 and S2 contain all the information for simulations. Through these simulations, the optimal thicknesses of each layer and interface properties were refined, further enhancing the device's performance. The mathematical model developed for this study incorporates the power generation of both the front and rear surfaces of the bifacial PSC, considering the effects of tilt angles (θ) and surface albedo (α) on the performance. The power generation from the front surface ($P_{front}(\theta)$) is given by:

$$P_{front}(\theta) = PCE_{front} \times [0.8D(\theta) + 0.2R(\theta)] \times I_0 \quad (1)$$

where, PCE_{front} is the front PCE from SCAPS-1D, $D(\theta)$ is the direct radiation factor:

$$D(\theta) = \sin(2\theta) e^{\left(\frac{-0.1(\theta-30)^2}{100}\right)} \quad (2)$$

$R(\theta)$ is the diffuse radiation factor:

$$R(\theta) = \frac{1 + \cos(\theta)}{2} \quad (3)$$

I_0 is the incident solar radiation, and θ is the tilt angle, which ranges from 0° to 45° .

For the rear surface power generation $P_{rear}(\theta, \alpha)$:

$$P_{rear}(\theta, \alpha) = PCE_{rear} \times \alpha \times [0.7D(\theta) + 0.3R(\theta)] \times I_0 \quad (4)$$

where, PCE_{rear} is the rear PCE from SCAPS-1D and α is surface albedo.

Total PCE:

$$PCE_{total} = \int EQE(\lambda) \times [\alpha_{front}(\lambda) + \alpha_{rear}(\lambda)] \times S(\lambda) d\lambda \quad (5)$$

where, $EQE(\lambda)$ is the external quantum efficiency, $\alpha_{front}(\lambda)$ and $\alpha_{rear}(\lambda)$ are the absorption coefficients from CASTEP for the front and rear surfaces, respectively, and $S(\lambda)$ is the solar spectrum.

$$Bifacial Factor = \left(\frac{Rear PCE_{eff}}{Front PCE_{eff}} \right) \times 100\% \quad (6)$$

Surface albedo values for different installation environments: fresh snow ($\alpha = 0.8$), concrete ($\alpha = 0.4$), grass ($\alpha = 0.25$), and water ($\alpha = 0.06$) [36].

While this study provides valuable insights through simulation-based analysis, it is important to acknowledge its limitations. The results are derived from numerical models and simulation tools, which, although highly predictive, may not fully capture real-world complexities such as material imperfections, environmental variability, and long-term degradation effects. Additionally, certain assumptions such as ideal interfaces and uniform material properties are inherent in the modeling process.

In the simulations, a defect density (N_t) of 10^{14} cm^{-3} was selected, as this value is widely reported for high-quality perovskite films and is experimentally achievable [18, 49]. This moderate defect density provides a realistic balance between carrier lifetime and recombination losses. Increasing N_t beyond this value is known to enhance trap-assisted recombination, which degrades carrier transport and reduces device performance [50]. Although further lowering N_t can artificially increase the simulated PCE, such low defect densities are difficult to realize experimentally and may lead to unrealistically optimistic efficiency predictions [51]. Therefore, $N_t = 10^{14} \text{ cm}^{-3}$ was chosen to ensure physical realism.

The series resistance (R_s) can be taken as $1 \Omega \cdot \text{cm}^2$ to represent efficient charge-transport with minimal ohmic losses [52, 53]. An increase in R_s is well known to degrade the fill factor and overall device efficiency due to increased resistive losses. While lowering R_s could slightly enhance the simulated PCE, excessively low values are often unrealistic for practical devices. Thus, $R_s = 1 \Omega \cdot \text{cm}^2$ should be selected as a realistic and experimentally achievable value.

Similarly, the shunt resistance (R_{sh}) should be set as $2000 \Omega \cdot \text{cm}^2$, as higher R_{sh} values suppress leakage currents and improve device performance [54, 55]. Reducing R_{sh} leads to increased shunt losses and degraded efficiency, whereas further increasing R_{sh} beyond this value produces only marginal performance improvements. Since $R_{sh} \approx 2000 \Omega \cdot \text{cm}^2$ is achievable in well-fabricated PSCs and adequately captures shunt-loss effects, it was chosen to maintain realistic device modeling.

In this study to verify the SCAPS-1D results, OghmaNano was used. OghmaNano, formerly known as General-Purpose Photovoltaic Device Model (GPVDM) is an advanced simulation suite created by Roderick C. I. Mackenzie at Imperial College London [56]. The software is open source and supports modeling of a wide range of devices, including organic and perovskite solar cells, OLEDs, and optical filters. Its electrical simulations are based on the self-consistent solution of Poisson's equation, the bipolar drift-diffusion equations, and the carrier continuity equations, enabling accurate prediction of key device characteristics, including $J-V$ curve [57]. Optical responses are computed through the transfer-matrix method, while numerical calculations are carried out using a finite-element-based solver.

3 | Results and Discussion

3.1 | DFT Calculations

3.1.1 | Electronic Results

As we can observe in Figure 2a–c, the DFT calculations determined the bandgap values for each layer in the bifacial PSC structure, confirming their energetic alignment and suitability for efficient solar absorption and charge-transport. The computed bandgaps were 3.184 eV for FTO, 1.64 eV for CdS, and 2.20 eV for Ca_3Ni_3 . These values ensure minimal energy loss during carrier transition across interfaces while maintaining strong light absorption in the active layer. The computed bandgaps of each material are in close agreement with the previously published data [58–60]. FTO, as a transparent conducting oxide with a wide bandgap, allows the majority of incident solar radiation to pass through without absorption, making it ideal for front illumination. CdS, with a relatively narrower bandgap, serves as an effective ETL due to its ability to extract and transport electrons from the absorber while blocking holes, thus reducing recombination. Ca_3Ni_3 , with a moderate bandgap of 2.20 eV, is well-suited as the absorber layer capable of harvesting a significant portion of the visible solar spectrum while also maintaining thermal and chemical stability. The favorable band alignment among these layers facilitates a strong built-in electric field, which aids in efficient separation and transport of photogenerated carriers. The Partial Density of States (PDOS) in Figure 2d of Ca_3Ni_3 shows that the valence band (below the Fermi level at 0 eV) is mainly dominated by I-5p and N-2p states, indicating strong hybridization between iodine and nitrogen orbitals. A sharp and intense peak around -7 to -8 eV is largely contributed by I-5p states, reflecting localized electronic states. Near the Fermi level, the density of states decreases significantly, suggesting a semiconducting nature with a small bandgap. In the conduction band (above 0 eV), Ca-4s states contribute more prominently, along with some hybridization with N-2p and I-5p states. These

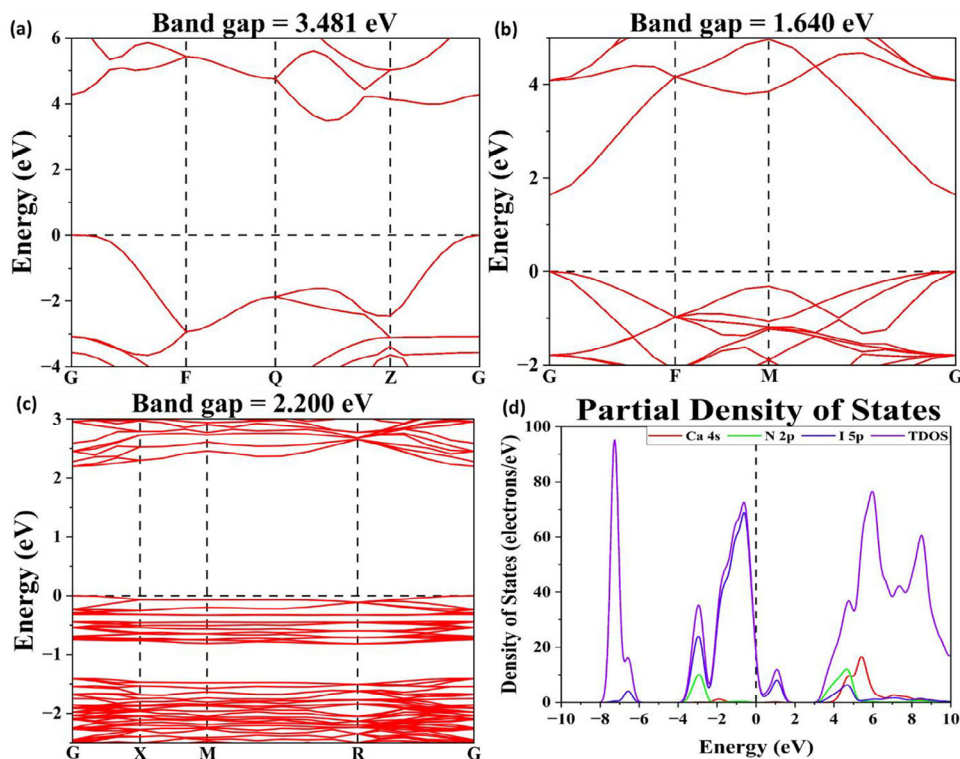


FIGURE 2 | Bandgap of (a) FTO, (b) CdS, (c) Ca₃Ni₃, and (d) PDOS of Ca₃Ni₃.

results collectively demonstrate the strategic selection of materials, contributing to the high PCE observed in the simulated device, as illustrated in Figure 2.

Figure 3a shows the crystal structure of cubic Ca₃Ni₃ is illustrated, where Ca (green), N (blue), and I (brown) atoms are arranged in a well-defined periodic lattice within the unit cell. The structure exhibits high symmetry characteristic of a cubic phase, with atoms occupying specific Wyckoff positions. The uniform spatial distribution of Ca, N, and I atoms highlights the structural stability and absence of significant distortions. Furthermore, the coordination environment suggests strong interactions between the constituent atoms, which are essential for understanding the bonding nature and overall stability. Figure 3b represents the 3D charge density distribution, which provides insight into the electronic interactions and bonding characteristics of Ca₃Ni₃. A significant accumulation of electron density is observed around the N and I atoms, indicating their higher electronegativity compared to Ca atoms. The relatively lower electron density around Ca suggests its role as an electron donor, supporting the ionic contribution to bonding. Additionally, the noticeable overlap of electron density between neighboring atoms indicates partial covalent character, confirming the presence of mixed ionic-covalent bonding.

3.1.2 | Mechanical Properties of Absorber Material (Ca₃Ni₃)

Table 1 presents the calculated elastic constants of cubic Ca₃Ni₃, including the longitudinal stiffness (C_{11}), transverse stiffness (C_{12}), and shear stiffness (C_{44}). The obtained values of $C_{11} = 58.421$ GPa, $C_{12} = 20.142$ GPa, and $C_{44} = 16.331$ GPa satisfy the

mechanical stability criteria for cubic crystals ($C_{11} > C_{12}$, $C_{11} + 2C_{12} > 0$, and $C_{44} > 0$), confirming the structural stability of the compound. Additionally, the positive value of $(C_{11} - C_{12}) = 38.279$ GPa indicates good resistance to shear deformation, while $(C_{11} + 2C_{12}) = 98.705$ GPa is related to the bulk modulus of the material.

Table 2 summarizes the derived mechanical properties of cubic Ca₃Ni₃, including bulk modulus (B), shear modulus (G), Young's modulus (E), Poisson's ratio (ν), Pugh's ratio (B/G), and Cauchy pressure ($C_{12} - C_{44}$). The calculated bulk modulus (32.902 GPa) and shear modulus (17.844 GPa) indicate moderate resistance to volume and shape deformation, respectively. The Young's modulus (45.783 GPa) reflects the material's stiffness, while the Poisson's ratio (0.282) suggests a balance between ionic and covalent bonding characteristics. Furthermore, the Pugh's ratio ($1.843 > 1.75$) and positive Cauchy pressure (3.811 GPa) confirm the ductile nature of Ca₃Ni₃.

3.1.3 | Thermodynamic Stability of the Absorber Material (Ca₃Ni₃)

As can be observed in Figure 3c, the variation of temperature as a function of molecular dynamics simulation steps demonstrates the thermal behavior of Ca₃Ni₃ under simulated conditions. The temperature exhibits fluctuations around an average equilibrium value, which is typical for systems in thermal equilibrium during molecular dynamics simulations. Despite these oscillations, no abrupt changes or instability are observed, indicating that the system maintains structural integrity throughout the simulation. The consistent temperature profile confirms that the material is thermally stable within the studied time scale and conditions.

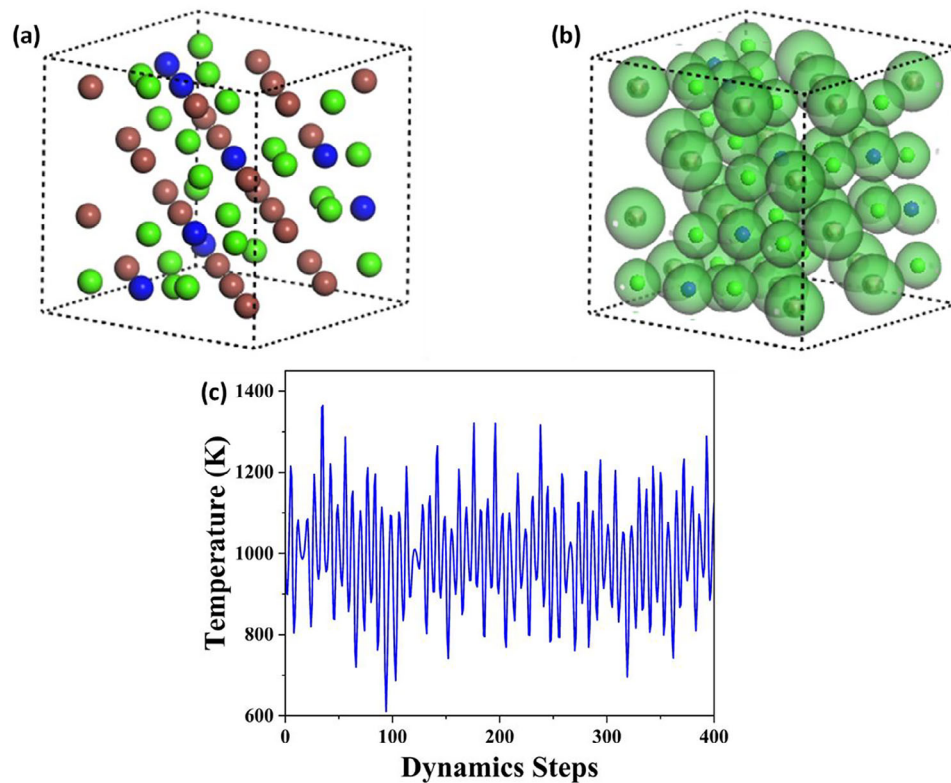


FIGURE 3 | (a) Crystal structure of cubic Ca_3NI_3 , where green, blue, and brown spheres represent Ca, N, and I atoms, respectively, within the unit cell. (b) 3D charge density distribution. (c) Temperature variation as a function of molecular dynamics simulation steps.

TABLE 1 | Computed Elastic Constants of Cubic Ca_3NI_3 .

Material name	Longitudinal stiffness (C_{11})	Transverse stiffness (C_{12})	Shear stiffness (C_{44})	$C_{11} - C_{12}$	$C_{11} + 2C_{12}$
Ca_3NI_3	58.421	20.142	16.331	38.279	98.705

TABLE 2 | Calculated Mechanical Properties of Cubic Ca_3NI_3 .

Material name	Bulk modulus (B)	Shear modulus (G)	Young's modulus (E)	Poisson's ratio (ν)	Pugh ratio (B/G)	Cauchy pressure ($C_{12} - C_{44}$)
Ca_3NI_3	32.902	17.844	45.783	0.282	1.843	3.811

This stability further supports the reliability of the structural and electronic properties obtained from the simulations.

3.1.4 | Optical Properties of Absorber Material (Ca_3NI_3)

Figure 4a shows that Ca_3NI_3 exhibits strong photon absorption, with a prominent peak at 26.92 eV and a high absorption coefficient of $3.31 \times 10^5 \text{ cm}^{-1}$, indicating efficient interband electronic transitions. This high-energy absorption originates from strong photon–electron interactions involving deeper valence states and higher conduction bands, which enable excitation of charge

carriers under ultraviolet (UV) illumination. Such extended UV absorption enhances the spectral utilization range of the solar cell, allowing the device to convert high-energy photons into useful charge carriers and thereby improve overall photoresponse [58].

As shown in Figure 4b, the real part of the dielectric function (ϵ_1) peaks at 2.75 eV with a value of 6.88, reflecting strong light–matter coupling and high polarizability of the electronic system in the visible range. A higher ϵ_1 value suggests that the material can effectively store electromagnetic energy, facilitating efficient exciton formation and dissociation under solar illumination. The energy loss function (Figure 4c) exhibits a peak at 28.46 eV

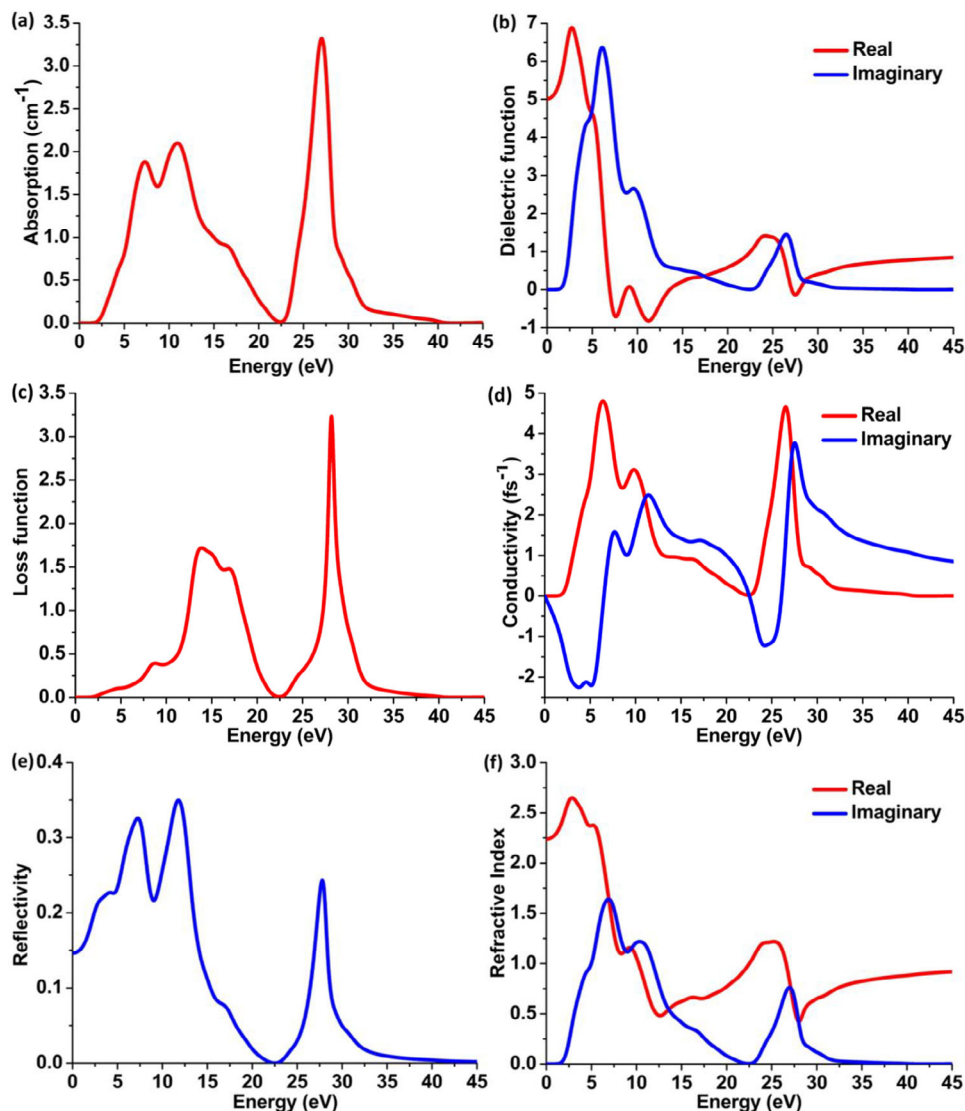


FIGURE 4 | Optical properties of Ca_3Ni_3 (a) absorption, (b) dielectric function, (c) loss function, (d) conductivity, (e) reflectivity, and (f) refractive index.

with a magnitude of 2.50, corresponding to the bulk plasmon resonance frequency. The relatively low plasmon energy loss indicates minimal damping of collective electron oscillations, which helps maintain carrier kinetic energy during transport and minimizes nonradiative recombination losses [61].

Furthermore, the optical conductivity spectrum (Figure 4d) peaks at 6.40 eV with a value of 4.81, revealing a high density of delocalized charge carriers and excellent carrier mobility. This signifies enhanced interband transition probability and reduced resistive losses, which are critical for efficient charge extraction and low series resistance within the device. The reflectivity spectrum in Figure 4e displays a modest peak of 0.35 at 11.93 eV, suggesting that only a small portion of incident photons are reflected. Such low reflectance implies effective optical impedance matching and better light trapping, both of which contribute to increased optical path length and photocurrent generation.

Lastly, the refractive index (n), shown in Figure 4f, peaks at 2.65 at 2.84 eV, signifying strong photon confinement and high

optical density in the visible range. A higher refractive index improves internal optical coupling within the absorber, reducing transmission losses and enhancing photon absorption probability [48]. Collectively, these optical characteristics demonstrate that Ca_3Ni_3 possesses superior dielectric response, strong light–matter interaction, and favorable charge-transport dynamics. Its ability to absorb both UV and visible photons, coupled with efficient exciton dissociation and minimal energy loss mechanisms, underscores its potential as a high-performance, eco-friendly absorber material for next-generation PSC.

3.1.5 | Optical Properties OF MgF_2 (ARC)

The reflectivity and refractive index characteristics of MgF_2 , as depicted in Figure 5a,b, highlight its effectiveness as an anti-reflective coating (ARC) in the visible and near-UV spectrum [62]. The material maintains low reflectivity across the lower energy range (0–50 eV), minimizing photon losses and enhancing light transmission. However, reflectivity peaks sharply at 62.50 eV with

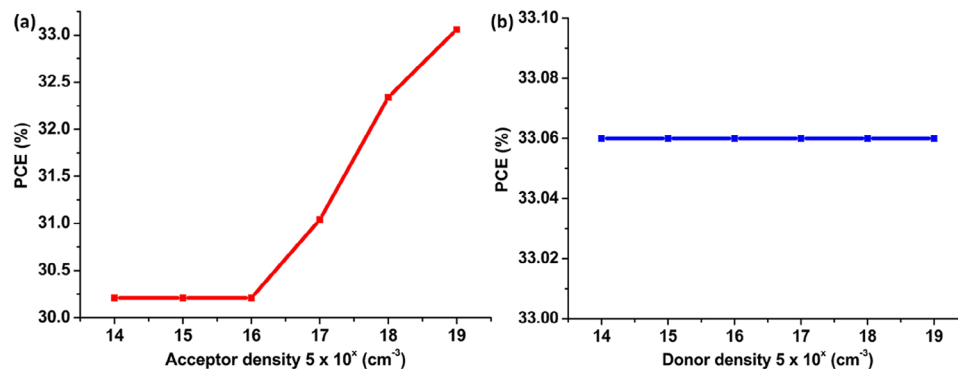


FIGURE 6 | Effect of (a) N_A and (b) N_D of Ca_3Ni_3 on PCE.

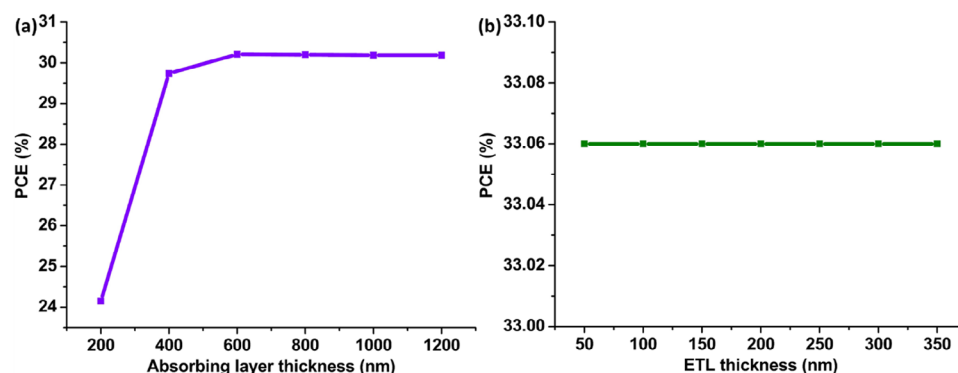


FIGURE 7 | Effect of (a) Ca_3Ni_3 and (b) ETL thickness on PCE.

recombination losses due to longer carrier transport paths and higher resistive losses, thereby limiting efficient charge extraction [67, 68].

3.2.5 | Variation of ETL Thickness and its Impact on PCE

Figure 7b and Table S6 show that the ETL thickness has a negligible effect on PCE, which remains constant at approximately 33.06% over the thickness range of 50 to 350 nm. This indicates that once the ETL is sufficiently thick to ensure efficient electron extraction and transport, further increases in thickness do not contribute to performance improvement. A thinner ETL of 50 nm is therefore preferable, as it minimizes series resistance and enhances charge collection efficiency. In contrast, thicker ETL layers may introduce additional optical losses by partially absorbing incident photons before they reach the Ca_3Ni_3 absorber [54]. Since the primary role of the ETL is to facilitate electron transport while blocking holes, an optimal ETL thickness of 50 nm is selected to achieve efficient charge extraction with minimal material usage and optical losses [69].

3.2.6 | Effect of Temperature and PV Performance of Device

Figure 8a and Table S7 illustrate the temperature-dependent performance of the proposed bifacial PSC, revealing a clear

downward trend in PCE with an increase in temperature. Specifically, the PCE decreases from approximately 33% at 300 K to around 30% at 400 K, underscoring the adverse impact of thermal effects on device performance. This decline can be attributed to a combination of interrelated physical mechanisms, predominantly arising from increased thermal agitation [13]. As the temperature increases, lattice vibrations intensify, which promotes phonon scattering and reduces charge carrier mobility. This leads to inefficient carrier transport and higher series resistance within the device. Moreover, elevated temperatures enhance the probability of non-radiative recombination via defect states. The thermally activated generation of defect states, particularly at grain boundaries and interfaces, facilitates trap-assisted Shockley–Read–Hall (SRH) recombination, which becomes a dominant loss mechanism under high-temperature conditions.

Thermal excitation also reduces the V_{oc} due to an increased intrinsic carrier concentration and a narrowing of the bandgap, both of which disrupt the delicate balance required for high voltage generation. In parallel, the structural and chemical stability of the perovskite absorber layer, Ca_3Ni_3 in this case, may be compromised under thermal stress. Prolonged exposure to elevated temperatures can lead to ion migration, phase instability, and degradation of interfacial layers, all of which further degrade charge separation and collection efficiency. The observed temperature sensitivity of the device not only highlights a fundamental challenge in PSC design but also emphasizes the critical importance of material selection, thermal interface engineering, and encapsulation techniques to mitigate heat-

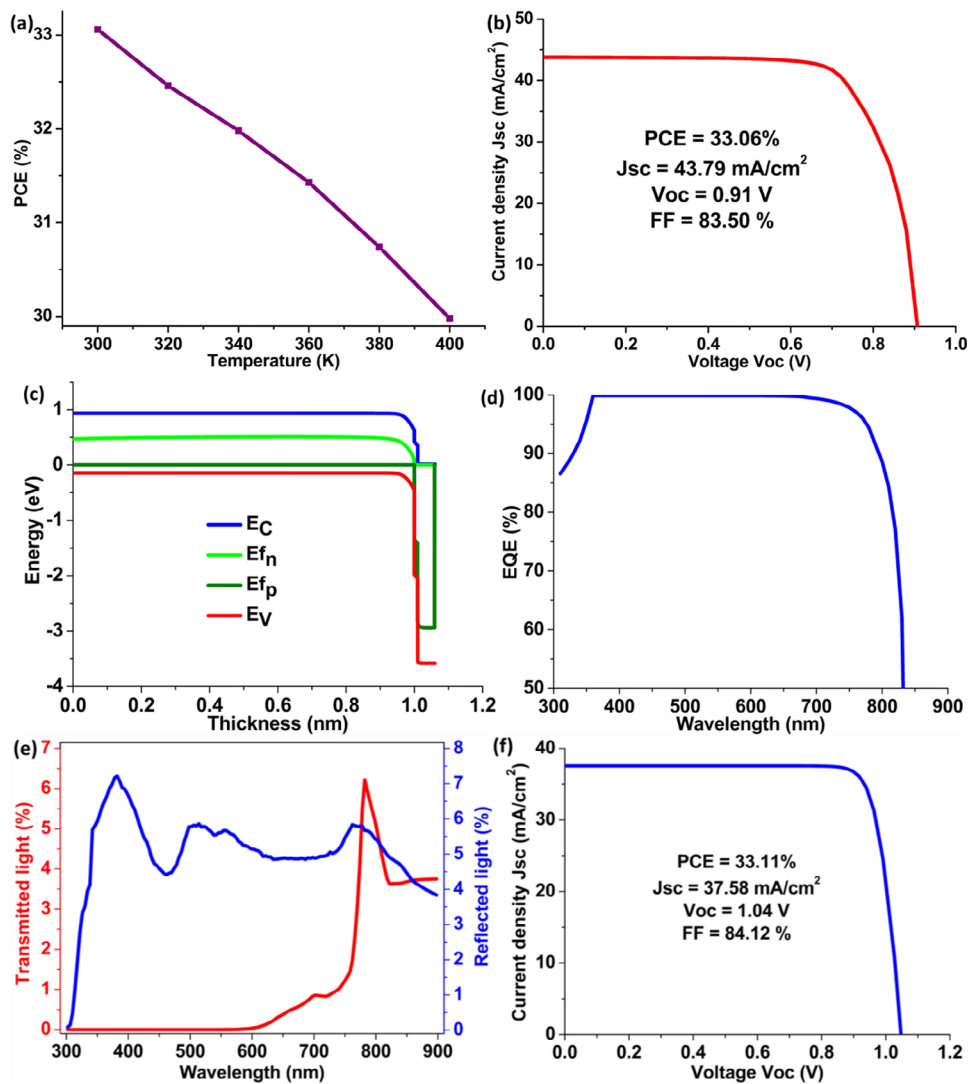


FIGURE 8 | (a) Effect of temperature on PCE, (b) J - V characteristics of proposed PSC by SCAPS-1D, (c) energy band diagram, (d) EQE curve of optimized PSC, (e) transmitted and reflected light curve, and (f) J - V characteristics of proposed PSC by OghmaNano.

induced degradation. Addressing these issues through advanced materials and thermal management strategies is essential to ensure long-term operational stability and commercial viability of PSCs, especially in real-world environments where devices are exposed to varying thermal conditions.

J - V (Current Density-Voltage) characteristics represent the electrical behavior of a solar cell, providing key parameters like J_{sc} , V_{oc} , FF, and PCE to assess performance. They are crucial for evaluating efficiency, optimizing materials, analyzing degradation, and comparing different photovoltaic technologies. Figure 8b presents the J - V characteristics of the proposed PSC, demonstrating a PCE of 33.06%, J_{sc} of 43.79 mA/cm^2 , V_{oc} of 0.91 V , and FF of 83.50% at 300 K.

Figure 8c presents the energy band diagram across the thickness of the FTO/CdS/ Ca_3Ni_2 device under steady-state illumination, clearly illustrating the conduction band (E_C), valence band (E_V), and quasi-Fermi levels for electrons (E_{F_n}) and holes (E_{F_p}). The selected layer configuration is intentionally designed to

achieve favorable band alignment that supports efficient charge separation, transport, and extraction [70].

At the FTO/CdS interface, the conduction band minimum of CdS is well aligned with that of FTO, forming a small positive conduction band offset. This alignment minimizes the energy barrier for electron transport from the absorber side toward the front contact while simultaneously blocking hole back-diffusion into the electron transport layer [71]. Such selective contact behavior is essential to suppress interfacial recombination losses and improve electron collection efficiency. Moving toward the CdS/ Ca_3Ni_2 interface, a clear staggered (type-II) band alignment is observed. The conduction band of Ca_3Ni_2 lies slightly higher than that of CdS, enabling photogenerated electrons in the absorber to be efficiently transferred into the CdS layer. At the same time, the valence band offset creates an energetic barrier for holes, confining them within the Ca_3Ni_2 absorber and preventing recombination at the CdS interface. This asymmetry in band offsets is a deliberate design choice, as it enhances carrier selectivity and promotes directional charge flow [72].

The pronounced band bending near the CdS/Ca₃NI₃ junction (around ~ 1.0 nm) indicates the presence of a strong built-in electric field arising from interfacial charge redistribution and doping gradients. This built-in field plays a critical role in driving photogenerated electrons and holes in opposite directions, thereby accelerating carrier separation and reducing recombination probability within the depletion region. Furthermore, the smooth spatial variation of the quasi-Fermi levels (E_{Fn} and E_{Fp}) across the absorber layer reflects efficient carrier transport under illumination and confirms that the device operates close to quasi-equilibrium conditions. The clear separation between E_{Fn} and E_{Fp} within Ca₃NI₃ indicates a strong photovoltage generation capability, directly contributing to the observed open-circuit voltage. Overall, the chosen FTO/CdS/Ca₃NI₃ architecture exhibits an energetically favorable band alignment that ensures low-resistance electron extraction through FTO, effective hole blocking at the CdS interface, strong built-in electric fields for carrier separation, and suppressed interfacial recombination. These features collectively justify the selected layer configuration and explain its suitability for achieving high photovoltaic performance [73, 74].

Figure 8d presents the external quantum efficiency (EQE) spectrum of the device, showing high and consistent photon-to-electron conversion across a broad wavelength range from approximately 350 to 800 nm. The EQE reaches nearly 100% in the visible region, indicating excellent light absorption and charge carrier collection efficiency. The high EQE values across most of the spectrum highlight the strong photo response and potential for high PCE in the device.

3.3 | Insights From Oghmanano Simulations

The optical response presented in Figure 8e, which includes the transmitted and reflected spectra of the optimized PSC, provides crucial evidence of the device's light-harvesting efficiency under AM1.5G illumination. The very low transmission and moderate reflection across the visible range indicate that most incident photons successfully enter and are absorbed by the active layer rather than being lost through optical escape. This behavior directly supports the high J_{sc} predicted by SCAPS-1D, as efficient photon absorption is the primary requirement for achieving a large photogeneration rate. The spectral profile also helps to confirm that the optical environment assumed in SCAPS aligns well with that reproduced in OghmaNano. Since OghmaNano employs the transfer-matrix method to calculate optical propagation, its agreement with SCAPS suggests that parasitic optical losses are sufficiently low, validating the feasibility of the SCAPS-derived J_{sc} value of 43.79 mA/cm².

The J - V characteristics shown in Figure 8f offer further confirmation of the electrical behavior predicted by SCAPS-1D. OghmaNano yields a PCE of 33.11%, with J_{sc} , V_{oc} , and FF closely matching the performance projected by the SCAPS model. This close correspondence between the two simulation platforms is of high scientific importance, as it demonstrates that the device behavior is not an artifact of a single numerical solver. SCAPS and OghmaNano rely on different mathematical approaches—SCAPS uses a 1D drift-diffusion scheme with simplified optical assumptions, while OghmaNano incorporates finite-element numerical

solvers and a more rigorous optical treatment. When two independent tools with different physical modeling strategies arrive at nearly identical results, it significantly strengthens confidence in the accuracy and physical realism of the simulated device. The small deviations in J_{sc} and V_{oc} between the simulators are expected due to differences in recombination modeling and optical treatment, but the overall alignment in PCE confirms that the device architecture is robust and well-optimized for operation at 300 K under AM1.5G illumination.

Together, the optical and electrical outputs from OghmaNano provide a strong scientific backbone to the SCAPS-1D results. They verify that the assumptions related to absorption, charge generation, recombination, and transport used in SCAPS are physically justified. This dual-model validation is an essential component of high-quality simulation studies, as it eliminates concerns that the high-performance metrics predicted by SCAPS may be artificially inflated. By demonstrating that OghmaNano reproduces nearly the same photovoltaic behavior, the study establishes that the proposed device design is genuinely capable of achieving high efficiency according to multiple independent modeling frameworks. This greatly enhances the credibility of the results and supports the overall reliability of the simulation-based predictions.

3.4 | Insights of Mathematical Model

3.4.1 | Effect of Tilt Angle and Albedo on PSC

Figure 9 illustrates the impact of tilt angle on the PCE of bifacial PSCs for different ground surfaces: 9(a) water (albedo = 0.06), 9(b) grass (albedo = 0.25), 9(c) concrete (albedo = 0.4), and 9(d) fresh snow (albedo = 0.8) [25, 36]. The bifacial PSCs rely on both direct sunlight and reflected light from the ground (albedo effect), and at a specific tilt angle, the combined contribution of these factors reaches its peak. The variations in PCE across different surfaces arise due to differences in reflectivity. Water (Figure 9a) exhibits the lowest rear-side contribution due to its minimal reflection, while fresh snow (Figure 9d) provides the highest enhancement due to its strong reflectance (albedo = 0.8), leading to a higher rear PCE. Grass (Figure 9b) and concrete (Figure 9c) exhibit moderate reflection levels, leading to an intermediate rear-side contribution.

The bell-shaped trend is a general characteristic in bifacial PSC performance with tilt angle variations due to changes in light absorption dynamics [75]. Initially, as the tilt angle increases from 0° to 35°, the solar cell aligns better with direct sunlight, and the rear-side contribution improves, leading to a gradual rise in PCE. At 35.20°, the incident solar irradiance and reflected light from the ground are optimally aligned, maximizing charge carrier generation and minimizing reflection losses. However, as the tilt angle exceeds 35.20°, self-shading effects become dominant, reducing the effective area exposed to direct sunlight. Additionally, at steeper angles, reflected light from the ground is less efficiently captured by the rear side, leading to a decline in PCE. This results in the observed symmetric bell curve in all cases, reinforcing that excessive tilting leads to a net reduction in efficiency due to increased optical and geometric losses.

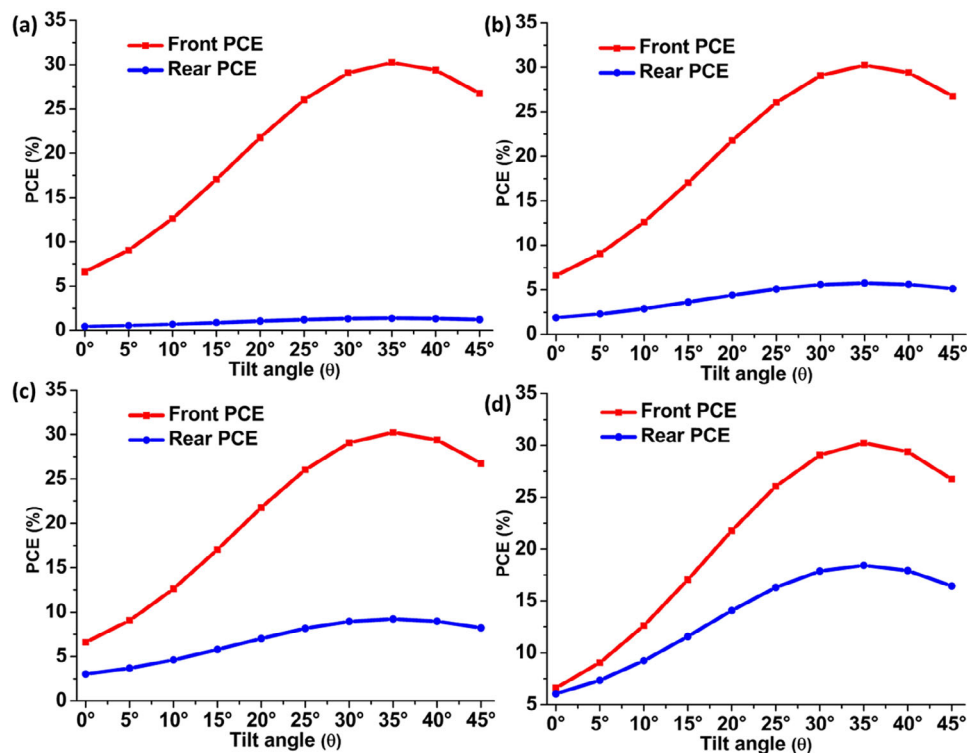


FIGURE 9 | Impact of tilt angle on PCE when the reflecting surface is (a) water (0.06), (b) grass (0.25), (c) concrete (0.4) and (d) fresh snow (0.8). The numerical values in brackets represent the albedo values for each surface.

The variation in the magnitude of rear PCE among different surfaces is due to differences in albedo values, affecting how much light is reflected back onto the solar cell [25]. For water (Figure 9a), the rear-side PCE remains nearly negligible, as water has a very low reflectivity (0.06), meaning minimal additional light is captured by the rear side. For grass (Figure 9b), a slight increase in rear PCE is seen, as grass reflects more light (albedo = 0.25), but its diffuse reflection pattern limits efficiency enhancement. Concrete (Figure 9c) further improves rear-side absorption, as it reflects more light (albedo = 0.4), leading to a more noticeable rear PCE contribution. The highest efficiency is observed for fresh snow (Figure 9d), as it has the highest albedo (0.8), significantly enhancing rear-side light absorption and improving total PCE. This explains why bifacial PSCs perform better in high-reflectivity environments such as snowy landscapes. The variation in PCE with tilt angle observed in Figure 9a–d is primarily governed by the interplay of optical interference, light absorption dynamics, and directional charge collection efficiency in the device structure $\text{MgF}_2/\text{FTO}/\text{CdS}/\text{Ca}_3\text{Ni}_3/\text{MgF}_2$. The front PCE shows a sharp increase with tilt angle, reaching a maximum around 30°–35°, before slightly declining. This behavior is attributed to enhanced light absorption due to increased optical path length and favorable antireflection conditions introduced by the MgF_2 layer at oblique angles [76]. At these tilt angles, constructive interference and reduced reflectance maximize photon absorption in the Ca_3Ni_3 perovskite absorber, leading to increased charge carrier generation and thus higher PCE. The rear PCE remains significantly lower than the front across all tilt angles but shows a moderate increase with angle in some cases. This is because light entering from the rear passes through the MgF_2 layer first and does not benefit as much from optimized optical interference and transparent conductive layers

like FTO [64]. Additionally, rear-side illumination often leads to less efficient carrier collection, particularly if the charge-selective layers are not optimized for bi-directional operation. In certain configurations (e.g., Figure 9c,d), the rear PCE shows noticeable improvement, which may be due to enhanced transparency or reflection management at the back interface, allowing some light recycling or internal scattering that aids absorption. Table 3 represents the comparison of PV parameters of similar bifacial PSCs.

4 | Feasibility of Device and Applications in Real-World

The proposed $\text{MgF}_2/\text{FTO}/\text{CdS}/\text{Ca}_3\text{Ni}_3/\text{MgF}_2$ bifacial PSC can be experimentally realized using standard thin-film deposition techniques. Each layer in the stack is compatible with scalable fabrication routes: MgF_2 can be deposited through electron beam evaporation [82], FTO can be grown using magnetron sputtering [83], and the CdS ETL can be formed using a chemical bath deposition process [84]. The Ca_3Ni_3 absorber layer is suitable for both spin-coating-based solution processing and thermal co-evaporation methods, allowing precise control over film thickness and crystallinity [85]. A rear MgF_2 layer can be deposited at low temperatures without damaging the perovskite, completing a fully manufacturable device stack. Since all these steps require no exotic or impractical equipment, the device architecture is experimentally feasible and compatible with large-area module fabrication.

With a high PCE of 33.06% and a bifacial factor of 76.245%, this optimized device structure is well-suited for real-world solar

TABLE 3 | Comparison of similar PSCs configuration on SCAPS-derived PV parameters. (* Simulated and # Experimental).

Solar cell configuration	Jsc (mA/cm ²)	Voc (V)	FF (%)	PCE (%)	Bifacial factor	Ref.
ITO/TiO ₂ /CH ₃ NH ₃ PbI ₃ /Cu ₂ O/ITO*	27.12	1.18	82.24	27.90	71.18	[32]
ITO/BCP/Ag/MoO ₃ (40 nm)*	17.73	1.05	72.45	13.49	71.24	[77]
ITO/BCP/Ag/MoO ₃ (60 nm)*	17.59	1.04	69.89	12.79	62.71	[77]
IZO/C60/Rb _{0.05} Cs _{0.05} MA _{0.05} FA _{0.85} Pb(I _{0.95} Br _{0.05}) ₃ /Back Contact*	36.39	1.174	73.0	31.2	96.15	[78]
FTO/ZnO/PCBM/MAPbI ₃ /Spiro/MoO ₃ /In ₂ O ₃ :H*	20.08	0.890	69.00	12.33	95.53	[79]
ITO/PEDOT:PSS/MAPbI ₃ /Fullerene/BCP/Ag/MoO ₃ *	17.45	1.05	72.73	13.49	71.24	[77]
ITO/PEDOT:PSS/MAPbI ₃ /PCBM/PEIE/Ag/ MoOx#	18.60	0.98	74	13.55	62.06	[80]
Glass/ITO/PEDOT:PSS/ Perovskite/PC61BM/BCP#	19.45	1.0	79.12	15.39	80.83	[81]
MgF ₂ /FTO/CdS/Ca ₃ NI ₃ /MgF ₂ *	43.79	0.91	83.50	33.06	76.245	This work

energy harvesting applications that demand maximum light utilization. The dual MgF₂ anti-reflective layers and bifacial design make the device effective in environments where both direct and reflected light contribute to power generation, such as rooftop installations, open-field solar farms, agrivoltaic systems, and building-integrated photovoltaics (BIPV). The mathematical tilt-angle model developed in this work further strengthens practical relevance, as it enables installers to determine the optimal mounting angle for maximum yearly output. This combination of high efficiency, manufacturable architecture, and environmental optimization makes the proposed bifacial PSC highly promising for next-generation solar power applications.

Despite the promising simulated performance, experimental realization of the MgF₂/FTO/CdS/Ca₃NI₃/MgF₂ bifacial architecture presents some practical challenges. In particular, achieving uniform, high-quality Ca₃NI₃ thin films with controlled stoichiometry and low defect density is critical, especially for large-area fabrication [86]. Interface quality between CdS and Ca₃NI₃ must be carefully engineered to suppress interfacial recombination, while precise thickness control and strong adhesion of MgF₂ layers on both sides are required to maintain optical transparency and bifacial functionality [87]. Additionally, effective encapsulation and environmental protection remain important considerations for stable outdoor operation.

5 | Conclusion

In this work, a high-efficiency bifacial perovskite solar cell with an MgF₂/FTO/CdS/Ca₃NI₃/MgF₂ architecture was systematically designed and optimized using an integrated framework combining DFT, SCAPS-1D simulations, OghmaNano modeling, and a mathematical tilt-angle analysis. This multi-scale approach establishes a direct connection between material-level properties, device optimization, numerical validation, and real-world operational performance. DFT calculations confirmed suitable structural, mechanical, electronic, thermodynamic, and optoelectronic properties of the absorber layer. SCAPS-1D optimization yielded excellent photovoltaic performance, achieving Jsc of 43.79 mA/cm², Voc of 0.91 V, a fill factor of 83.50%, and

PCE of 33.06% at 300 K. The bifacial configuration delivered a rear-side PCE of 25.21% and a bifacial factor of 76.245%, demonstrating efficient utilization of both direct and reflected light. Parametric analyses identified an optimal Ca₃NI₃ thickness of 500 nm and a CdS ETL thickness of 50 nm, balancing optical absorption, charge-transport, and recombination losses. Temperature-dependent studies revealed a gradual decline in efficiency at elevated temperatures, highlighting the importance of thermal considerations. The reliability of the SCAPS results was independently validated using OghmaNano simulations, which showed closely matching J–V characteristics and photovoltaics metrics. Finally, the mathematical model revealed an optimal tilt angle of approximately 35°, at which bifacial performance is maximized. High-albedo surfaces, such as fresh snow, significantly enhanced rear-side power generation, while low-albedo environments contributed minimally. Overall, this study presents a robust and physically consistent design strategy for high-efficiency, bifacial PSC and provides valuable guidance for their practical implementation under realistic outdoor conditions.

Author Contributions

Anshul Aggarwal: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis. **Manasvi Raj:** Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis. **Sajal Batra:** Software Investigation. **Aditya Kushwaha:** Software. **Neeraj Goel:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

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